CIFAR-10 Image Classification

Udacity AI Mastery Capstone · Deep Learning Systems

Report Overview

This project trains a convolutional neural network (CNN) to classify 32 × 32 color images from CIFAR-10 into ten object categories. It compares two Adam learning rates over eight epochs to measure how step size affects learning progress and held-out classification performance. The dataset is publicly available at https://www.cs.toronto.edu/~kriz/cifar.html (Krizhevsky, 2009).

Dataset and Task Description

CIFAR-10 contains 60,000 real RGB images, with 6,000 per class (Krizhevsky, 2009). I held out 500 images per class from the official training data with seed 42, leaving 45,000 training and 5,000 validation images. The official test set has 10,000 images, with 1,000 per class. The loader reads the official binary archive, converts pixels to floating point in [0, 1], and uses integer class labels. The ten labeled categories support supervised single-label classification.

Model Architecture and Design Decisions

The CNN has three convolutional blocks with 32, 64, and 128 output channels. Each padded 3 × 3 convolution learns local features while preserving spatial dimensions. ReLU adds nonlinearity, and 2 × 2 max pooling reduces the spatial grid from 32 × 32 to 4 × 4 across the three blocks. Increasing the channel count gives later blocks more feature maps as the grid shrinks. This design follows the general principle of combining learned local filters with spatial subsampling (LeCun et al., 1998).

A linear layer maps the 2,048 flattened features to ten logits. Multiclass cross-entropy compares those logits with the true class, and the largest logit determines the prediction (Goodfellow et al., 2016). Adam adapts per-parameter updates using estimates of gradient moments (Kingma & Ba, 2015). The held-out results show that this architecture learned useful features under the baseline configuration, although the cat-to-dog errors show where classification remains weak.

Experimental Comparison

The baseline used a learning rate of 0.001 and the experiment used 0.0003. Adam's learning rate scales parameter updates, making it a meaningful variable for comparing convergence within a fixed training budget (Kingma & Ba, 2015). Both runs used the same initial weights, seed, data split, minibatch order, batch size of 128, preprocessing, architecture, loss, Adam defaults apart from learning rate, and eight epochs. Test metrics use the final epoch of each run.

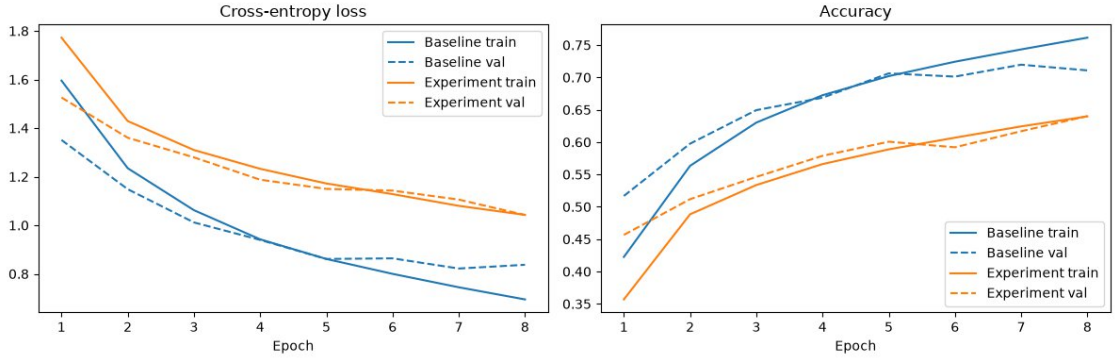
Results and Interpretation

Metric	Baseline (0.001)	Experiment (0.0003)
Test accuracy	70.40%	63.18%
Test macro F1	0.7008	0.6303
Final train minus validation accuracy	+5.06 pp	−0.05 pp
Lowest validation loss	0.822 at epoch 7	1.042 at epoch 8

The baseline led by 7.22 percentage points on test accuracy. Accuracy measures the overall fraction of correct predictions. Macro F1 averages the per-class F1 scores equally, incorporating both precision and recall (Sokolova & Lapalme, 2009). Both metrics favored the baseline on the balanced test set.

Figure 1

Training and Validation Loss and Accuracy by Epoch



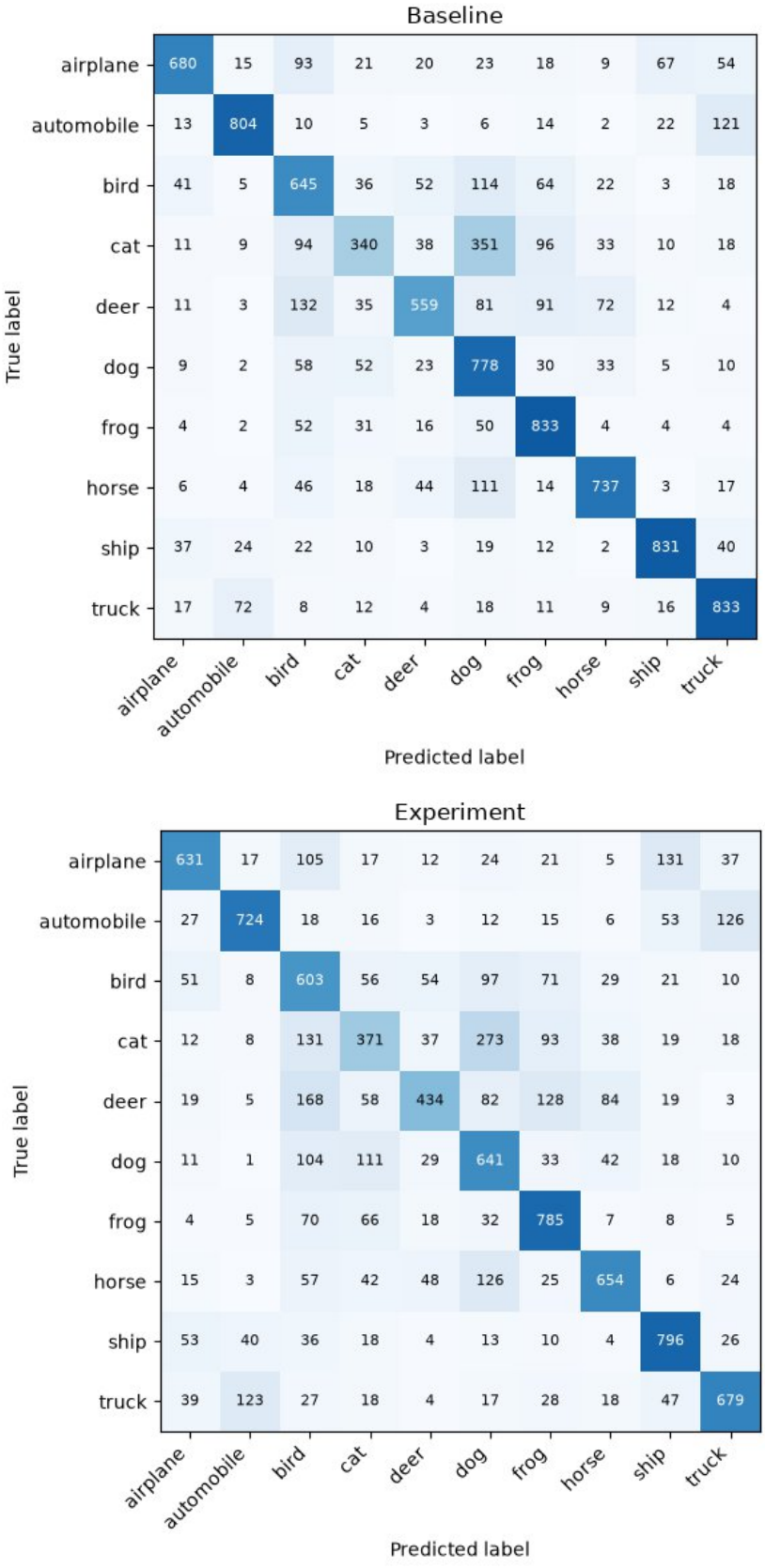
From epoch 7 to 8, baseline training loss fell from 0.745 to 0.695 while validation loss rose from 0.822 to 0.837 and validation accuracy slipped from 71.96% to 71.08%. This pattern is consistent with possible early overfitting (Goodfellow et al., 2016). Validation loss also rose slightly from epoch 5 to 6 (0.861 to 0.864) before recovering, so a one-epoch rise could reflect normal variation.

The experiment's validation loss fell from 1.106 to 1.042 between epochs 7 and 8. Its lower accuracy and still-improving loss are consistent with slower convergence within this budget. More epochs would be needed to determine whether its performance catches up.

The baseline predicted dog for 351 of 1,000 true cats (35.1%), including test image index 0, and correctly labeled only 340 cats (34.0% recall, the lowest of any class). The confusion was asymmetric. Only 52 true dogs were labeled cat, and dog recall was 77.8%. The baseline predicted dog 1,551 times against 1,000 true dogs, drawing errors from birds (114), horses (111), and deer (81) as well as cats. This shows a tendency to overpredict dog across several animal classes in this run (Figure 2).

Figure 2

Test Confusion Matrices for Both Learning Rates



Note. Rows show true labels. Columns show predictions.

Limitations and Risks

The fixed eight-epoch budget measures performance at equal training effort but may favor the rate that converges sooner. A single seed gives no estimate of run-to-run variability. Training omitted augmentation, so robustness to shifts, flips, and lighting changes remains unmeasured. The low resolution obscures fine details, and benchmark results do not establish reliability on photographs from other cameras or settings.

Ethical and Responsible Use

CIFAR-10 was derived from Tiny Images (Krizhevsky, 2009). Birhane and Prabhu (2021) documented offensive labels and other harms in large image datasets, including the parent collection that was subsequently withdrawn. That history calls for checking dataset provenance and label quality before reuse. A ten-class classifier will also force unfamiliar inputs into one of its known categories. In a downstream wildlife or shelter workflow, a wrong prediction could misroute an image, so uncertain or out-of-scope cases should go to human review.

Future Improvements

Run both learning rates across several paired seeds and report the mean and variability. In a separate experiment, give both configurations the same larger epoch ceiling and validation-loss early-stopping rule, then evaluate their best validation checkpoints. Test random crops and horizontal flips in a separate controlled comparison and track per-class recall and predicted label counts to see whether cat recall improves and excess dog predictions decrease.

References

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